

INFORMATION RETRIEVAL FOR EVERYDAY FOOD QUESTIONS

EvidenceAtlas

Food



Provides Answers to Your Questions
Using Scientific Papers.

A working search and comparison tool for everyday readers, dietitians and researchers

Name: Adil Muhammed

Roll no: 2410110018

WHAT WE BUILT, IN FIVE STEPS

01

Collect real literature

Find nutrition papers in PubMed and Europe PMC.

02

Keep the paper structure

Separate abstracts, methods, results and other sections.

03

Build and combine searches

Match exact terms and meaning, then rank candidate papers.

04

Make the evidence inspectable

Open source passages, compare studies and save an investigation.

05

Measure what the search returns

Review ranked papers and calculate Precision@3, @5 and @10.

950 searchable papers

494 full texts

37,975 passages

01 / COLLECT AND ORGANIZE

Start with the literature.

The collection was built from scientific records, then turned into searchable sections.

973

unique records

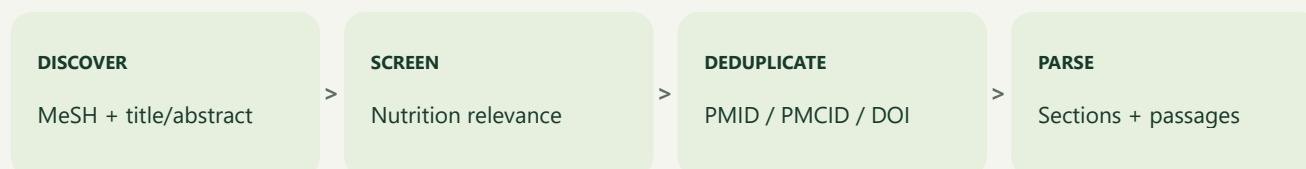
494

ingested full texts

12

nutrition topic groups

We searched by scientific subject headings and by words in titles and abstracts. The topic plan covered foods, nutrients, meal timing and preparation: for example caffeine and sleep, sodium and blood pressure, and cooking and vitamin retention.

THE PATH FROM A RECORD TO EVIDENCE

When an open-access full text was available, the JATS XML was downloaded and parsed. Each passage keeps its parent paper, section name and position. This lets a result open at the relevant evidence while the reader checks the surrounding methods or discussion.

FULL TEXT

More than the abstract

494 records contain ingested full text. Methods, results, discussion and tables can be inspected where the source supplies them.

ABSTRACT ONLY

A visible limit

473 records have abstracts only; six contain metadata only. These availability labels stay visible rather than implying a complete paper was read.

Raw responses and content hashes are retained. The first 77-record pilot included 42 full texts and informed later collection growth.

02 / BUILD THE SEARCH

Find the words. Find the meaning.

Different retrieval methods solve different parts of the same search problem.

LEXICAL SEARCH

Exact terms and structure

TF-IDF and BM25 use an explicit index of words, counts and positions. Section-aware BM25 can give a match in results more weight than a match in background text.

SEMANTIC SEARCH

Related wording

MiniLM turns each passage into a 384-number vector. A question is encoded the same way, so related wording can match even when the exact words differ.

All 37,975 passage vectors are complete. The model reads up to 256 tokens per passage; 2,723 longer passages were truncated for encoding. Their complete text is still stored and searchable through the lexical index.

HYBRID + RELEVANCE RANKING

Bring candidates together, then look again.

Reciprocal Rank Fusion combines lexical and semantic rankings. A MiniLM cross-encoder then reorders up to 20 candidates using the question, paper title and a selected passage.

What All Techniques were Implemented

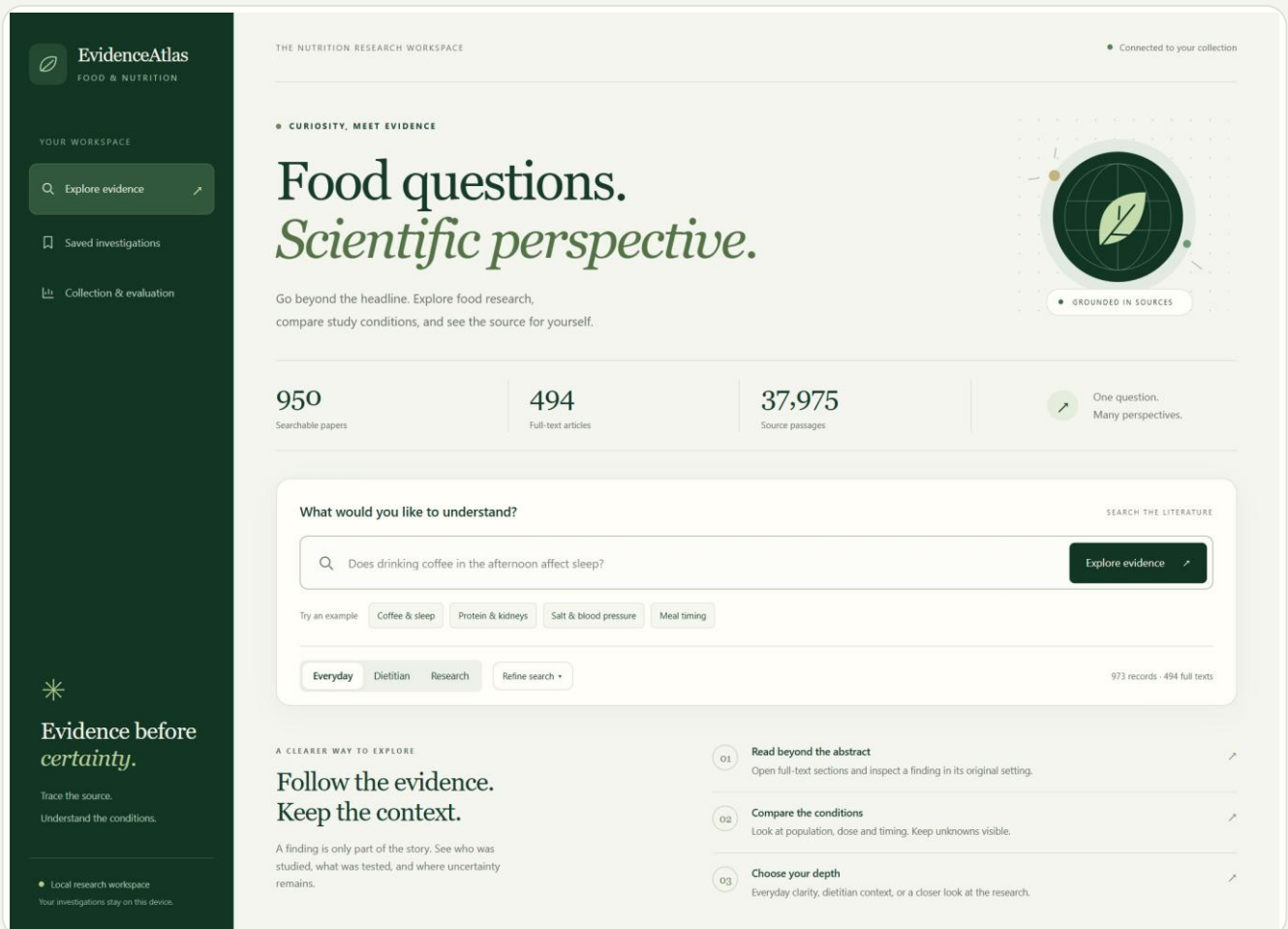
Hence in total 6 search techniques can be selected and used. These are Section-Aware BM25, Flat BM, Flat idf-cosine, Semantic (vector embedding), Hybrid(Uses BM and semantic), Hybrid + relevance ranking which uses vector model on top of hybrid that reranks the documents based on understanding. This was the most successful one out of all and has future promise for improvements.

The reranker sees a selected passage, not the entire paper. Its score estimates relevance to the question; it does not determine whether a scientific claim is true.

03 / USE THE PRODUCT

A search result you can open.

The interface carries the question through to source inspection and study comparison.



ACTUAL LOCAL INTERFACE / PHTHALO GREEN #123524

Everyday

A concise way to search and open relevant evidence.

Dietitian

More study context and side-by-side paper comparison.

Research

Retrieval traces and experimental model diagnostics.

A result card shows the paper and evidence availability. Opening it reveals source passages and surrounding sections. Readers can compare up to four papers, save notes with an investigation, and export evidence for later use.

FOOD QUESTIONS. SCIENTIFIC SOURCES.

04 / TEST THE RETRIEVAL

What counts as a useful result?

Precision measures the proportion of retrieved papers that help address the question.

Precision@k = relevant papers in the first k / k

For example: eight relevant papers among the first ten gives Precision@10 = 80%.

We ran six everyday questions with hybrid plus relevance ranking and inspected the top ten papers for each. Four query rewrites were also tried. Two matched BM25 searches provide a small baseline comparison. The corpus and retrieval settings were kept fixed.

Grade	How the paper is treated
2 - relevant	Directly useful for part of the question; relevant null findings count too.
1 - related	Background, or a mismatch in food, preparation or outcome.
0 - off-topic	Does not address the exposure and outcome being searched.

Only grade 2 counts toward precision. Judgments use titles, abstracts and selected source sections. They assess retrieval relevance rather than study validity or certainty.

A CONCRETE QUERY REFINEMENT

"Does eating salt raise blood pressure?" was rewritten as "Dietary sodium reduction blood pressure clinical trials". Precision@10 rose from 60% to 80%, while Precision@3 fell from 100% to 66.7%. The added detail found more useful papers overall but did not improve every rank.

The milk and egg rewrites were screened but not retained. All attempted queries and their review status are saved.

05 / READ THE RESULTS

Useful results, at three depths.

Five selected presentation examples using hybrid plus relevance ranking.

Query	P@3	P@5	P@10
Does coffee affect sleep?	100.0%	100.0%	90.0%
Is eating a late night meal bad?	100.0%	80.0%	80.0%
Does boiling vegetables remove vitamins?	66.7%	60.0%	80.0%
Dietary sodium reduction blood pressure clinical trials	66.7%	80.0%	80.0%
Does drinking milk help keep bones strong?	100.0%	80.0%	50.0%
Selected-example mean	86.7%	80.0%	76.0%

These examples were chosen after inspecting results. The original six-question set, including the weaker egg query, averaged 88.9%, 76.7% and 63.3% at ranks 3, 5 and 10. The selected table is a demonstration, not a general accuracy estimate.

SAME QUERIES, DIFFERENT METHOD

Two-query paired comparison	P@3	P@5	P@10
Flat BM25	100.0%	70.0%	75.0%
Hybrid + relevance ranking	83.3%	90.0%	85.0%

Pair: coffee/sleep and the refined sodium query. Hybrid did better at ranks 5 and 10; BM25 did better at rank 3. Two queries cannot establish a general winner.
FOOD QUESTIONS. SCIENTIFIC SOURCES.

06 / DELIVERY AND NEXT STEPS

A working system, with room to grow.

The project connects acquisition, full-text retrieval and source inspection in one local workflow.

WHAT IS WORKING

Search to comparison

Real literature ingestion, section and positional indexes, passage embeddings, hybrid retrieval, reranking, source navigation and saved investigations.

WHAT WAS CHECKED

Code and interface

23 core/dense tests and the TypeScript check passed in the delivery check. Earlier browser checks exercised search, source viewing, comparison, saving and export.

This belongs in Track 6 as it is domain specific (Research papers relating to food only) and rely on structures, zones, metadata, citations etc. This is hence a search engine build for professional domain that dietitians, researchers in this field can use as well as casual people. This project can also slightly come under Track 2 as there are various techniques incorporated including vectoring and does hybrid as well where it merges the scores.

RUN AND REPRODUCE

```
python -m evidenceatlas.cli serve --port 8765
```

Run from the app folder with EAF_ROOT set to the data directory, then open <http://127.0.0.1:8765/>.
README.md includes full installation instructions.

```
python scripts/evaluate report.py
```

The evaluation script recalculates every reported score from saved rankings and labels. Papers, indexes, models and caches stay on E:. The measured project uses about 4.1 GB; acquisition reserves 40 GB free and supports further growth. Models run locally on CPU without a paid inference API.

07 / LIMITATIONS AND POSSIBLE IMPROVEMENTS

Current limits. Clear next steps.

The system is useful today, but broader coverage and stronger ranking are the clearest paths forward.

CURRENT LIMITATION

More papers mean more embedding work.

The collection has 973 unique records, but only 494 have ingested full text. Adding more research papers should improve coverage. Every new passage also needs indexing and a vector embedding, so a much larger semantic index will take additional time and CPU compute.

MODEL IMPROVEMENT

A stronger model above hybrid retrieval.

Hybrid + relevance ranking is the strongest current setup, but the reranker only reviews up to 20 candidates and sees a selected passage rather than the whole paper. A stronger reranker on top of the hybrid stage could separate close or ambiguous results more reliably.

LOCAL LLM OPTION

Keep the intelligence local.

A local LLM could sit after hybrid retrieval and reranking to compare the strongest evidence, rerank borderline papers, or produce a short evidence-grounded summary. This keeps the workflow local and can avoid a paid inference API. Any generated answer should still point back to the retrieved passages and citations.

AI DECLARATION: AI was used throughout the project including the UI, this report, fetching relevant documents etc

Name: Adil Muhammed

Roll no: 2410110018